DATABASE MONITORING

A Comprehensive Guide to Performance, Health & Incident Response

Database Systems & Reliability Engineering

OBJECTIVE

Equip engineering and operations teams with a structured framework to proactively monitor database systems, detect early failure signals, and respond effectively to incidents. Minimizing customer impact and ensuring data integrity.

- Define what to monitor across performance, health, capacity, and error signals
- Provide clear alert thresholds and incident response playbooks for each metric
- Map monitoring signals to the tools teams already use in production environments

AGENDA

01
Overview
Why database monitoring matters

02
Critical Monitoring Elements
Replication, connections, IOPS, queries, throughput, space & errors

03
Performance Monitoring
Signals, collection methods & incident actions

04
Health Monitoring
Connection errors & infrastructure health

05
Monitoring Tools
Oracle Enterprise Manager, Splunk, Amazon CloudWatch

06
Key Takeaways
Summary & best practices

OVERVIEW

Why Database Monitoring Matters

- Database incidents → customer-facing outages
- Degraded performance impacts end users directly
- Data integrity risks from undetected lag or errors
- Proactive monitoring prevents reactive firefighting

This Guide Covers

01 Critical database monitoring elements

02 Monitoring examples & incident response patterns

03 Common tools used to collect and alert on database telemetry

Effective monitoring focuses on: Performance · Health · Capacity · Error Signals

CRITICAL MONITORING ELEMENTS

Replication Lag

Delay between primary & replica sync

Open Connections

Active client sessions on the DB server

IOPS

Read/write operations on storage subsystem

Long-Running Queries

Queries exceeding execution time thresholds

Throughput

Transactions & queries processed per second

Space Usage

Storage utilization of volumes & tablespaces

Errors

DB-level errors: connection, query, disk

1. REPLICATION LAG

What It Measures

Delay between transaction commit on primary and application on replica. Measured in seconds, transactions, or Log sequence numbers (LSN differences).

Root Causes

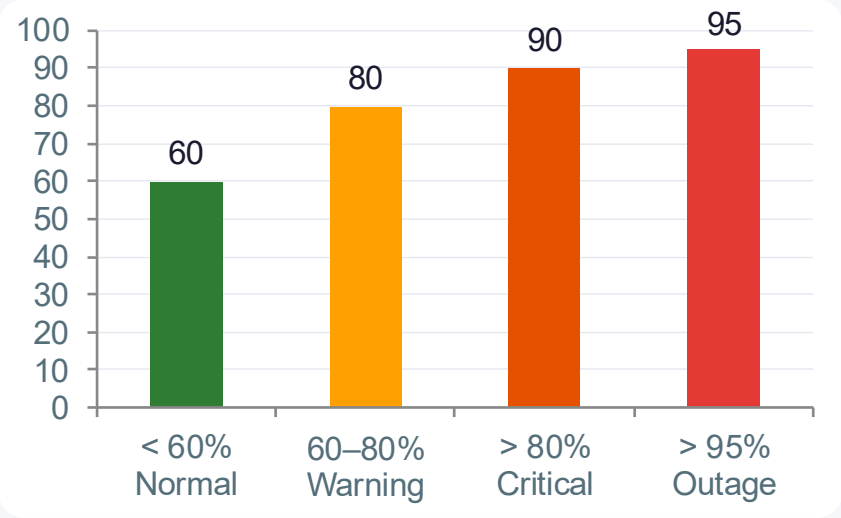
- Disk I/O bottlenecks
- Network latency between nodes
- Long-running transactions
- Replica CPU saturation
- Write-heavy workloads

Alert Thresholds

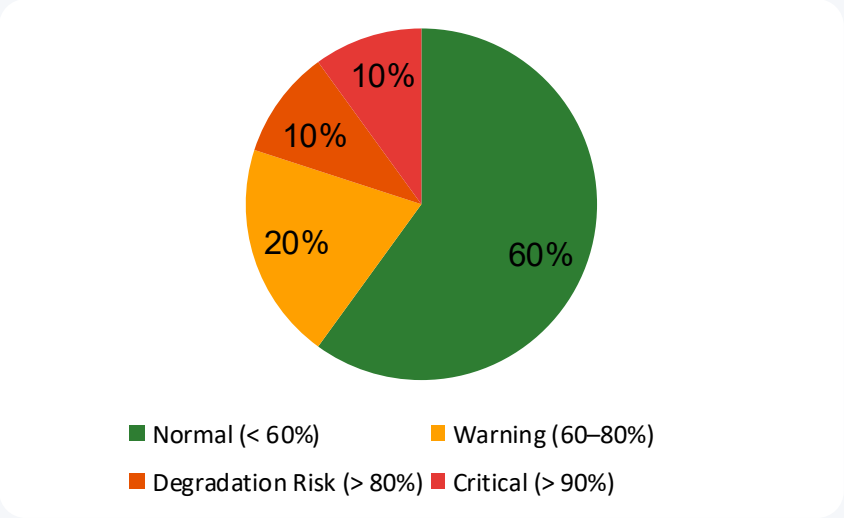
Lag Duration	Severity
< 5 seconds	✅ Normal
5–30 seconds	⚠️ Warning
> 30–60 seconds	🔴 Critical
Minutes+	Potential Replication Failure

2. OPEN CONNECTIONS & IOPS

Open Connections – Alert Thresholds



IOPS – Storage Utilization Thresholds



Connections: Incident Actions

Identify top connection sources · Check query performance · Investigate connection pool config · Kill runaway sessions if needed · Scale DB instance if sustained

IOPS: Incident Actions

Check disk latency metrics · Identify high I/O queries · Verify backup / maintenance jobs · Increase provisioned IOPS · Query optimization & index creation

4. EXPENSIVE / LONG-RUNNING QUERIES

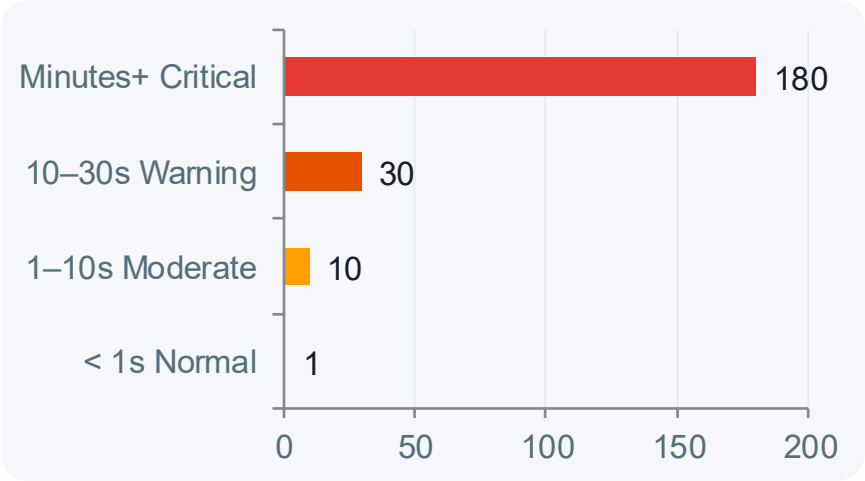
Why It Matters

- Blocks other transactions
- Exhausts CPU or memory
- Increases lock contention
- Degrades entire database performance

Root Causes

- Missing indexes
- Inefficient joins
- Large result sets
- Poor query plans
- Lock contention

Alert Thresholds by Query Duration

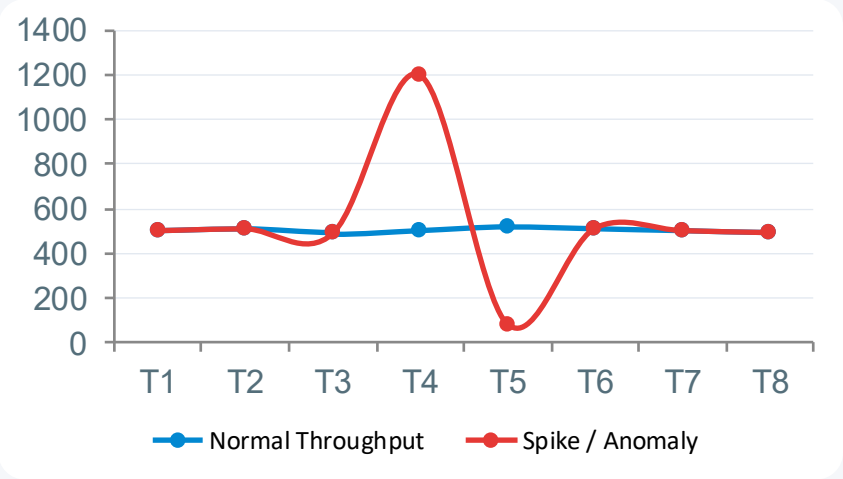


Incident Response Actions

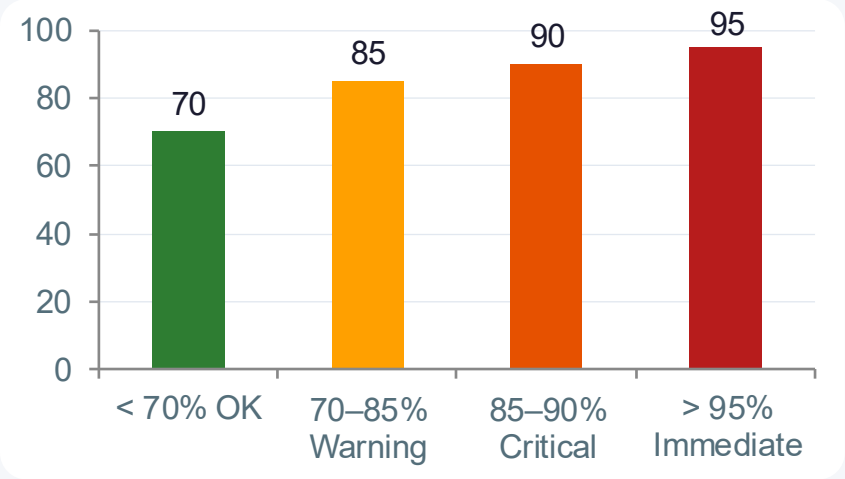
Identify fingerprint · Review execution plan · Check lock contention · Kill query if impacting system · Escalate to DB engineering

5. THROUGHPUT & SPACE USAGE

DB Throughput – QPS/TPS Patterns



Storage Usage – Alert Thresholds



Throughput Patterns

Sudden Spike: Traffic surge or runaway process

Sudden Drop: Service degradation or outage

Gradual Rise: Capacity pressure building

Space Usage: Root Causes

Log Growth: Unarchived transaction logs

Data Ingestion: Large bulk inserts

Temp Tables: Uncontrolled temporary growth

7. ERRORS & HEALTH MONITORING

Error Monitoring

Tracks database-level error events:

- Connection failures
- Query failures
- Replication errors
- Disk errors

Alerts trigger on:

- Error rate spikes
- Specific critical error codes
- Repeated failure patterns

Response: Review logs → Identify pattern → Correlate with deployments → Validate DB availability

Health Monitoring

Connection Errors

Normal: Very low error rate **Abnormal:** Sudden spike in failures

↳ Verify DB availability, check auth & network

VM / Infrastructure

Normal: CPU < 70%, stable memory **Abnormal:** CPU saturation, memory exhaustion, disk spikes

↳ Scale VM, restart instance, investigate leaks

SAMPLE MONITORING TOOLS

Oracle Enterprise Manager

Deep Oracle Diagnostics

Key Capabilities

- Real-time DB performance monitoring
- Query performance analysis
- Storage monitoring
- Replication status tracking

Best Used For

Deep Oracle database diagnostics, performance tuning, capacity planning

Splunk

Log Analysis & Observability

Key Capabilities

- Database log ingestion
- Query performance analysis
- Error pattern detection
- Incident correlation

Best Used For

Error analysis, root cause investigation, cross-system correlation

Amazon CloudWatch

AWS Cloud-Native Monitoring

Key Capabilities

- Metrics for RDS & Aurora
- Storage and IOPS monitoring
- Replication lag tracking
- Alarm-based alerting via SNS

Best Used For

Infrastructure-level DB monitoring, cloud-native alerting

TOOL ALERT WORKFLOWS

Oracle Enterprise Manager

1. Metric threshold triggered
2. Alert generated in OEM console
3. Incident ticket created
4. DBA investigates query or resource issue

Splunk

1. Log anomaly detected
2. Alert rule triggered
3. Incident escalated to responders
4. Root cause investigation begins

Amazon CloudWatch

1. Metric threshold exceeded
2. CloudWatch alarm triggers
3. Notification sent via SNS
4. Incident created in ITSM system

KEY TAKEAWAYS

1

Monitor 7 Critical Signals

Replication Lag, Open Connections, IOPS, Query Performance, Throughput, Space Usage, and Errors form the complete monitoring baseline.

2

Threshold-Based Alerting Saves Incidents

Define clear Normal / Warning / Critical thresholds for each metric to catch degradation before it causes outages.

3

Fast Incident Response Requires Runbooks

Each metric has defined response actions; identify, investigate, remediate, and escalate to minimize MTTR.

4

Use the Right Tool for the Right Signal

Oracle EM for deep diagnostics, Splunk for log correlation, CloudWatch for cloud-native infrastructure monitoring.

5

Health + Performance = Full Coverage

Combine performance monitoring (slowdowns, bottlenecks) with health monitoring (errors, VM resources) for complete observability.